

KOUSTUBH SHAH

Lead Backend Engineer | .NET Core | Microservices | System Design

Udaipur, Rajasthan, India | koustubhshah19@gmail.com | +91-8209560369 | [linkedin.com/in/koustubhshah](https://www.linkedin.com/in/koustubhshah)

PROFESSIONAL SUMMARY

Lead Backend Engineer with 7 years of experience building and scaling production systems in .NET Core, microservices, and cloud-native architecture. Currently leading backend development at Kadel Labs on WumDrop — a Walmart-affiliated last-mile delivery platform — engineering geofencing systems, automated dispatch algorithms, 3D vehicle-packing logic, ElasticSearch indexing, AI-powered proof-of-delivery validation, and self-healing microservices across a 200K+ line codebase. Proven ability to lead Agile teams end-to-end: sprint planning, stakeholder communication, cross-functional delivery, and technical documentation. Strong problem-solving track record with a consistent pattern of identifying gaps proactively and delivering reliable, scalable solutions.

CORE TECHNICAL SKILLS

Languages: C#, .NET Core (.NET 3.1 / 6 / 10), HTML5, CSS3

Backend & APIs: Web API, SignalR, RESTful APIs, Microservices Architecture, API Gateway, Webhooks, JWT, OAuth2, IdentityServer4, RBAC

Databases & Search: PostgreSQL, MySQL, SQL Server, Entity Framework Core, ElasticSearch (indexing, archival), Query Optimization, Stored Procedures

Messaging & Caching: RabbitMQ, Redis (Pub/Sub, Caching), Asynchronous Processing, Cron Jobs, Scheduled Reports

Cloud & DevOps: Docker, Kubernetes (K8s), Terraform, CI/CD Pipelines, GCP, GitLab, GitHub, Vercel

AI & Vision: Google Vision API, Media AI (image validation scoring), AI-powered productivity tooling

Algorithms & System Design: Google OR-Tools, Vehicle Routing Problem (VRP), Dijkstra's Algorithm, 3D Bin Packing, Geofencing, Distributed Systems, Self-healing Architecture

Frontend: React, Angular, Blazor, Tailwind CSS

Agile & Leadership: Agile/Scrum, Sprint Planning, Daily Standups, JIRA, Confluence, Technical Documentation, Architecture Diagrams, Stakeholder Management, Cross-functional Team Leadership, Mentoring

Tools & Platforms: JetBrains Rider, Postman, DBeaver, Apache Superset, Salesforce, GraphHopper API, Google OR-Tools, Google Maps API, MapsProxy

PROFESSIONAL EXPERIENCE

Kadel Labs Pvt. Ltd. — Lead Engineer

Jan 2025 – Present

Udaipur, Rajasthan, India | WumDrop — Walmart-affiliated Last-Mile Delivery Platform, South Africa

Promoted to Lead Engineer based on delivery of core platform algorithms and reporting infrastructure during Senior role.

- Engineered territory re-alignment and **geofencing system** handling **33,000+ polygons** with API response times of **~0.2 seconds**, operating within a **200,000+ line .NET codebase**.
- Built an **ElasticSearch indexer microservice** to index deliveries, jobs, users, and drivers for fast dashboard search — with automated **30-day archival logic** to move completed records to archive and maintain index performance.
- Engineered an **Auto Healer microservice** — a self-healing reconciliation system that detects missing or unsynced records in the dashboard (e.g., caused by indexer failures) and automatically restores them, eliminating the need for manual intervention during partial system failures.
- Integrated **Google Vision API** via a **Media AI** service to automatically score and validate driver-uploaded Proof of Delivery (POD) images — determining whether photo evidence meets delivery confirmation standards without manual review.
- Built and managed a **MapsProxy VM** acting as a unified gateway for all Google Maps API and Google Vision API calls, centralizing key management, rate limiting, and cost control across the platform.
- Built a full **Learning Management System (LMS)** backend in coordination with the UI team, supporting structured onboarding and training workflows for platform users.
- Implemented **audit logging** across platform services to capture and store a complete trail of user actions for compliance, debugging, and operational visibility.
- Integrated **Redis caching** to reduce repeated database reads and cut API response times; implemented **cron jobs** and **Webhook integrations** for automated scheduling and real-time third-party event notifications.
- Hardened platform security with **JWT/OAuth2 authentication** and **role-based access control (RBAC)** across all client-facing APIs.
- Led **Agile/Scrum** delivery across 3 concurrent enterprise projects — ran daily standups, sprint planning, retrospectives, and weekly stakeholder updates; managed backlog and risk tracking in **JIRA**.

- Built the platform's entire **technical documentation from scratch in Confluence** — including architecture diagrams, microservice interaction maps, and onboarding guides for a codebase that previously had no documentation.
- Mentored junior developers and trainees through code reviews, architecture walkthroughs, and hands-on training on **.NET Core and AI tooling** workflows.

Kadel Labs Pvt. Ltd. — Senior Software Developer (.NET)

Jun 2023 – Dec 2024

Udaipur, Rajasthan, India | WumDrop — Walmart-affiliated Last-Mile Delivery Platform, South Africa

- Built the core backend for **WumDrop** using **.NET Core, Entity Framework Core, and PostgreSQL** — covering order ingestion, delivery management, and driver operations; deployed via **GitLab CI/CD** on **Docker** and **Kubernetes**.
- Designed and built an **automated driver auto-assignment algorithm** — replacing a fully manual dispatch process — using zip-code clustering, proximity scoring, cost-per-job parameters, and OTD (On-Time Delivery) ranking; recognized with a **salary increment and bonus** for this contribution.
- Developed a **3D bin-packing algorithm** to optimize parcel-to-vehicle allocation: calculates volumetric fit (L×B×H) per parcel against available vehicle sizes, maximizes fill-rate to increase driver earnings per trip, and cascades overflow parcels to the next available vehicle.
- Set up **Apache Superset from scratch** as the platform's reporting layer — built data models, operational dashboards, and scheduled report delivery for client-facing analytics.
- Integrated **Google OR-Tools** (Vehicle Routing Problem + Dijkstra's Algorithm) with **GraphHopper API** and Google Distance Matrix API to generate optimized multi-stop delivery routes.
- Developed automated delivery grouping using **zip-code clustering and distance calculation**, replacing manual dispatcher grouping and reducing operational overhead.
- Implemented **RabbitMQ** message queuing for asynchronous processing; designed a **packet-size optimization model** to stabilize platform performance during peak delivery windows.
- Led backend sub-team through code reviews and **Postman**-based API testing across **RabbitMQ**-connected microservices.

Independent Software Developer — Freelance

Jan 2021 – May 2023

Self-Employed | Banswara, Rajasthan

- Designed, built, and deployed a full-stack **e-commerce platform** end-to-end using **.NET Core** — covering product catalog, inventory management, and order tracking.
- Integrated **Razorpay** payment gateway for secure online transactions and built an admin dashboard for order and inventory management.
- Owned full deployment pipeline: server setup, database optimization, and performance tuning to maintain platform stability under real traffic.

Infosys Pvt. Ltd. — Systems Engineer

Oct 2019 – Jan 2021

Nagpur, Maharashtra

- Built a full-stack web application using **Angular, HTML5, CSS3, JavaScript** on the frontend and **.NET Core** on the backend, leading the frontend development workstream within a 2-person team.
- Managed **SQL Server** databases including stored procedures, query optimization, and schema design for business-critical reporting.
- Provided technical support on a **Salesforce**-based client engagement — generated scrub reports and tracked resolution of open tickets to SLA.

EDUCATION

B.Tech in Software Engineering — SRM Institute of Science and Technology, Chennai

May 2019

CGPA: 8.32 / 10